

Volume 2, Book 2
**Part 3—Plans, Specifications,
and Estimates**

Section 3.01 General

A. Purpose

This part of the manual prescribes and standardizes procedures, arrangements, and format for the preparation of plans, specifications, and estimates (PS&E) for highway construction projects.

To develop, process, and design projects from the feasibility study stage to the tender document stage requires many engineering, technical, and clerical manhours of work. It is necessary, therefore, that certain minimum procedures and standards be prescribed to expedite and control the many complex and interrelated stages.

A set of plans, a set of specifications, a cost estimate, an equipment list and time determination, a Bill of Quantities, and a Design Report shall be prepared by the Engineer during the final design and specifications phase for each proposed construction project. These documents, together with the Right-of-Way and Utility Relocation Plans previously prepared under 2-2.01 and 2.02, constitute the PS&E assembly. The PS&E documents are intended to be complementary and to describe and provide for a complete project.

B. Definitions

1. Plans

The project drawings, consisting of all sheets and drawings necessary to show the location, character, dimensions, and details of the work.

2. Standard Drawings

Drawings issued by the MOC as HDM-4.

3. Specifications

A general term applied to all directions, provisions, and requirements that are obligatory on the contractor in his performance of the work.

4. General Specifications

The current book of General Specifications for Road and Bridge Construction of the MOC.

5. Supplemental Specifications

Additions or revisions to the General Specifications that are adopted after issuance of the printed book.

6. Special Specifications

Additions and revisions to the General Specifications and Supplemental Specifications concerning conditions peculiar to an individual project. Special Specifications describe new or seldom used ideas and unusual conditions.

7. Estimate

The estimated total cost for construction of the project at the time of receipt of tenders.

8. Design Report

A report prepared by the Engineer and containing all engineering data, analyses, and schedules of work required for the design and construction of a proposed highway project (see 2-3.05).

9. Form of Tender

The required MOC form on which the Tenderer submits his offer to perform the work at the price quoted.

10. Contract Time

The number of working days or calendar days allowed for completion of the work. Contract time is also known as the contract period.

11. Equipment List

A theoretical list, developed by the Engineer, of equipment that could be used to construct a project.

12. Bill of Quantities

A tabulation developed by the Engineer of all items and quantities of work to be included in a contract.

C. Background

The standardization of the PS&E for highway and bridge construction projects is essential in order to provide uniformity and consistency, facilitate the review and approval of the documents by the Ministry, facilitate design and construction, develop complete construction cost estimates, and provide a uniform basis for tenders.

The Engineer shall develop plans and specifications that conform to the procedures, arrangements, and format set forth in these guidelines. Plans and specifications shall be prepared using SI units of measurement.

The Engineer shall use the standard drawings and General and Supplemental Specifications whenever possible. The use of these documents will simplify and facilitate the interpretation and use of the project plans and specifications by contractors and others performing the construction operations and furnishing material and equipment. The use of the standard drawings and General and Supplemental Specifications also reduces a duplication of effort in producing individual sets of plans and specifications that involve features and provisions that are a part of the majority of construction projects.

3.01 C

The Engineer shall prepare a project estimate that states the expected cost of the project. The estimate shall be prepared on a standard format estimate sheet (see Figures 3.04.1 and 3.04.2). The Engineer shall prepare a Design Report to be included with the PS&E assembly.

The PS&E assembly developed by the Engineer will require approval by the MOC before the project can be advertised for tenders. The Engineer shall submit six copies of the assembly for the approval process. The PS&E doc-

uments shall not be printed in final form until after approval has been obtained.

The PS&E assembly submitted for approval shall include all plan drawings that are not standard drawings and all specifications that are not a part of the General Specifications. Standard drawings shall not be included in the assembly but shall be included by reference on the plan sheet showing the list of drawings.

After approval, copies of the PS&E documents shall be furnished to the MOC in accordance with sections 2-3.07 and 2-3.08.